

Presentation 3

What did each of the team members do?

Note that multiple team members can work together on the same task. In that case, you may write the same task for all of them. For example,

Aidan: Helped make mandatory changes to presentation 3 base model from Krish (using entropy regularization and gated attention implementations from other teams' presentation 2). Generated new data splits 1-5 that, together, contain all 35 benign cases. Ran model with mandatory changes on these new splits and reported results.

Relevant code is in

/projects/e32998/STAT390_Winter2026/Team5/Code/Code6_NewSplits_27Feb_Aidan.

Emma: Created a new module that finds patch-level classifications and creates various plots and extracts images of patches for analysis.

Fauzan: Stepped out the math for motivating keeping slice=level hierarchy and understanding resolution as a hyperparameter

Paris: Assisted with mandatory changes (specifically, removing dropout from models.py) and worked on [Optuna hyperparameter tuning](#). I edited my tuning script ([optuna_tuner_paris.py](#)) to not shuffle validation data and to include the other hyperparameters. I ran the optuna training over 20 hours with 20 trials and 30 epochs, and the results are [here](#). I then trained the model on these results, which can be found in /projects/e32998/MIL_training/final_runs/run_20260301_125350/.

Ronaldo: Used Emma's new module that finds patch-level classifications and creates various and extracts images of patches for analysis for the optuna model that Paris built.

How does it help the project?

Aidan: We were able to gain a fuller understanding of how the current version of the model is performing on the benign cases. By looking at the benign cases that are still being misclassified, we can think about what further changes to make. We also see that the model is sometimes performing worse on high-grade cases than benign, so we can take that into account as well.

Emma and Ronaldo : Getting patch-level predictions helps us understand why the model makes mistakes at the case level. Instead of only seeing whether a whole case was classified correctly or not, we can see how each small image patch was classified. By reporting the percentage of patches predicted in each class, we can tell whether the

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error came from most of the tissue looking suspicious or just a small number of misleading patches.

Fauzan: Gives direction to how we should proceed with model design

Paris: It helped us to understand better how Optuna tuning can help us to improve the model. It at the very least helped us to understand what optuna training will require in terms of trials and epochs for future attempts.

Issues faced that haven't been resolved:

Optuna hyperparameter trained model is overfitting the data, and not able to learn the patterns well enough in the data.

We noticed that the code updates that fixed attention analysis significantly decreased model performance, so we should fix this before proceeding further.

Proposed next steps:

References